Defect Triage, Root-Cause and Retest Case Study

Document ID	CASE-UAT-002
Revision	1.0
Status	ILLUSTRATIVE — NOT APPROVED FOR OPERATIONS
Audience	QA/UAT Lead, Procurement SME, Engineering and Business Owner
Information classification	Synthetic portfolio exercise; no employer or customer data

Scenario: A fictional business consolidates five legacy datasets into a new ERP staging environment. All systems, transactions, personnel and results in this document are synthetic.

1. Scenario overview

During a synthetic UAT cycle, test TC-002 loaded supplier VND-0042 from a legacy export with approval_status=APPROVED but an empty approval_reference. The AtlasERP staging display showed Active and selectable. This violated FR-003 and mapping MAP-02.

2. Initial evidence and expected behavior

Evidence attribute	Observed	Expected
Fixture ID	SUP-APPROVAL-MISSING-01	Fixture version locked
Source vendor	VND-0042	Preserve source lineage
Approval reference	Blank	Required to activate
Target status	Active / selectable	Blocked / not selectable
Build / mapping	stg-0.9.7 / map-v1	Record in test run
Test result	FAIL; DEF-002 opened	Block and create EXC-003

3. Containment and communication

- UAT Lead marks TC-002 FAIL, creates DEF-002, links requirement FR-003 and MAP-02, and classifies potential S1 due to possible unauthorized purchase.
- Procurement confirms whether documentary approval exists and clarifies business meaning; Engineering isolates the faulty mapping rule.
- Test Lead recommends blocking the affected vendor subset in staging until an approved and retested correction is available.
- Business Owner documents that the UAT exit decision is NO-GO while the issue remains open.

4. Root-cause hypothesis and verification

The example defect is caused by a transformation rule that maps APPROVED to Active without checking a nonblank valid approval reference. This is a synthetic hypothesis supported by a failing negative test; a real root-cause conclusion would require code review, configuration inspection and owner validation.

To verify, compare mapping-v1 transformation logic with MAP-02; run a minimal dataset containing a valid approved vendor, an approval-status-only vendor, and a clearly rejected vendor. Verify expected status of all three.

5. Correction, retest and regression plan

Action ID	Owner	Evidence required	Status in sample
ACT-01	Engineering	Updated transformation with	Proposed

		approval evidence gate	
ACT-02	Procurement	Signed acceptance of approval-source field semantics	Proposed
ACT-03	QA/UAT	TC-002 re-execution on build stg-0.9.8	Not executed
ACT-04	QA/UAT	Regression TC-001, TC-003, TC-008	Not executed
ACT-05	Business Owner	Exit-criteria assessment after evidence review	Pending

6. Suggested retest procedure

- Use an unchanged, versioned version of SUP-APPROVAL-MISSING-01 as input.
- Capture the corrected build identifier, mapping hash or version, date and tester.
- Confirm VND-0042 is blocked and cannot be selected in the material creation flow.
- Confirm EXC-003 records the source and target values, discrepancy, owner and next action.
- Run nearby positive and negative cases to ensure the fix did not block all approved suppliers.
- Attach results and update DEF-002 only after evidence passes and required review occurs.

7. Mock status and professional writing note

At publication, DEF-002 remains OPEN, actions are proposed, and the example business decision is NO-GO. This deliberate open state teaches evidence discipline: a writer must not label corrective actions complete without execution and sign-off. No case shown here is an actual customer or employer incident.

8. Usability review questions

- Can a new reviewer identify the expected/actual gap without asking the original tester?
- Can the relevant business owner find which field they must verify?
- Can the developer identify a deterministic negative test?
- Does the packet separate observations, hypotheses, actions, and closure evidence?

REGULATORY / TECHNICAL REVIEW ADDENDUM — November 2026

Independent technical writing portfolio enhancement. Additional validation gates and examples are illustrative, not authorized production instructions. Regulated applicability varies by product, process, record and organization.

Reproduction and evidence preservation

Reproduce using fictitious approved and inactive vendor identifiers, capture build/version, test account, input fixture, mapping version, and timestamp. Preserve the original failed result and a redacted export or screenshot within an access-controlled test record, not a publicly posted live environment image.

Root-cause hypotheses include stale crosswalk, incorrectly mapped status flag, source approval history not transferred, and insufficient UAT boundary coverage. Validate hypotheses against system evidence rather than assigning blame prematurely.

Fix, retest, regression and owner communication

Assigned developer corrects transform or validation logic under change control; IT confirms build provenance; business owner verifies intended decision rule; QA or compliance assesses whether the fix affects governed records.

Execute negative and positive cases: inactive supplier blocked, approved supplier allowed, unknown supplier quarantined, duplicate supplier detected, alternate source feed mapped accurately. Mark closure only after reviewer sees actual run ID and observed result; the project contains fictional demonstration outcomes only.

Communicate a status update stating impact, workaround, owner, next evidence milestone and whether deployment is blocked. This provides the technical writer with a reusable stakeholder-facing communication example.

Document-specific reference basis and status

ISO/IEC/IEEE 26514:2022 information for software users (full URL in portfolio review report)

IEC/IEEE 82079-1:2019 instructions for use (full URL in portfolio review report)

FDA Part 11 scope and application guidance (full URL in portfolio review report)

W3C WCAG 2.2 color use guidance (full URL in portfolio review report)

Status: technical/editorial review against publicly available references completed at a portfolio-example level; no licensed-standard full-text conformance audit, live environment testing, site QA approval or legal certification performed.

Appendix: November 2026 standards and documentation applicability review

PUBLIC SYNTHETIC PORTFOLIO SAMPLE — This appendix records source-checking and editorial scope boundaries. It does not make the fictional organization regulated, certify a system, or replace actual company policies and competent quality/legal review.

ERP / UAT regulatory applicability and software test-evidence review

- Confirm each source-to-target mapping has an owner, approved transformation rule, exception disposition and independent tie-out. Synthetic test failures must remain OPEN until a documented rerun passes.
- Separate functional UAT from regulated computerized-system validation: business acceptance is not evidence of 21 CFR Part 11 compliance or FDA-required validation.
- 21 CFR Part 11 may become relevant only if the mapped environment maintains qualifying electronic records/signatures within predicate-rule scope; evaluate access, audit trail, record retention and copies when applicable.
- For medical-device production/QMS software, FDA's February 2026 Computer Software Assurance final guidance provides risk-based recommendations and QMSR became effective February 2, 2026; do not impose these onto a general commercial SAP migration.
- NIST access control, change management and audit logging offer voluntary security reference points for privileged ERP access, defect fix approvals and evidence retention.
- Technical writing QA: include acceptance criteria, reproducible steps, observed versus expected result, severity basis, defect owner, approval and version; never infer that an unresolved hypothetical defect passed.

Reference links and recheck notes

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/computer-software-assurance-production-and-quality-management-system-software

https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final

Verified against published authoritative source pages as of November 2, 2026. No proprietary standards were audited clause-by-clause. Actual application must be assessed based on product, system use, region, and record type. Proposed procedures have not been executed in a production environment.

Independent second-pass document review | November 2, 2026

REVIEWED PORTFOLIO DEMONSTRATION — this appendix records a document-specific editorial and source-scope check. The main procedure and forms are fictional and not approved for production, executed testing or regulated use. Original source versions are retained as editable files in this portfolio.

A. Document-specific findings and checks

Review objective	2026 reviewer correction / expectation	Verification evidence
Reproducibility	Provide stable sample ID, preconditions, expected/actual result, environment, revision and screenshots/log reference placeholders.	Reviewer can reconstruct the synthetic failure without confidential company data.
Cause vs effect	Separate observed wrong vendor status from suspected business-rule defect.	Remediation includes preventing downstream PO approval until disposition.
Closed-loop test	Proposed patch -> unit/integration test -> regression/UAT retest -> business acceptance.	Do not describe remediation as closed without test output and owner acknowledgment.

B. Applicable regulatory / guidance boundaries

Review objective	2026 reviewer correction / expectation	Verification evidence
Scope	Fictional commercial supplier master/ERP migration, not a validated medical-device QMS or clinical EDC.	Do not label SAP-like mock UAT as an FDA validation or actual system access test.
Evidence	Requirements -> source mapping -> expected/actual result -> defect -> retest -> acceptance owner.	Use synthetic IDs, revisioned mappings and traceable test executions.
Conditional FDA applicability	Part 11 if predicate electronic records; 2026 QMSR and device CSA only if used for in-scope device production/QMS.	A business test pass never independently proves GxP compliance or authorized production deployment.

C. Evidence and sign-off boundaries

Status: Editorial second-pass reviewed; technical procedure not executed; company-specific SOP approval not obtained.

All tests, screenshots, users, suppliers, dates, IDs, incident observations, performance metrics and signatures in the demonstration are synthetic.

Next practitioner action: confirm source applicability for the actual jurisdiction and product; conduct SME/user testing and accessibility review; update controlled revision and approval fields.

No color, watermark, revision number, PDF export or reference link by itself demonstrates regulatory compliance.

D. Authoritative primary-source references

FDA 21 CFR Part 11 scope: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/part-11-electronic-records-electronic-signatures-scope-and-application

FDA device QMSR Feb 2026: https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr

FDA device CSA Feb 2026: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/computer-software-assurance-production-and-quality-management-system-software

US Access Board Section 508: https://www.access-board.gov/ict/about/